

ECO102: Introduction to Macroeconomics

Lecture 4: Monitoring Jobs and Inflation Contd.

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Price Level, Inflation & Deflation

- ***Price level*** is the average level of prices and the value of money

This average level of prices is usually calculated for a “basket of goods and services”

The market basket is a representative selection of items that everyday consumers buy.

The Bureau of Labor Statistics (BLS) constructs the market basket using more than 80,000 items

Changes in the price level can impact consumer behavior, investment decisions, and overall economic growth

Price Level, Inflation & Deflation

- A persistent rising price level is called ***inflation***
- A persistent falling price level is called ***deflation***
- Low, steady, and anticipated inflation or deflation isn't a problem, but an unexpected burst of inflation or period of deflation brings big problems and costs

Why?

- Redistribution of Income
 - Redistribution of Wealth
 - Lowers Real GDP and Employment
 - Diverts Resources from Production
- In the extreme case an inflation rate of 50% or higher in a month grinds the economy to a halt and causes the society to collapse, this extreme case of inflation is called ***hyperinflation***

CPI and RPI

- The Consumer Price Index (CPI) and the Retail Price Index (RPI) are two ways to measure the average price level of an economy for a fixed basket of goods.
- The CPI and RPI is defined to equal 100 for ***a reference base year***
- Currently, The US CPI considers 1982-1984 to be the reference base period. For Bangladesh the reference base year is 2021-2022
- Meaning:

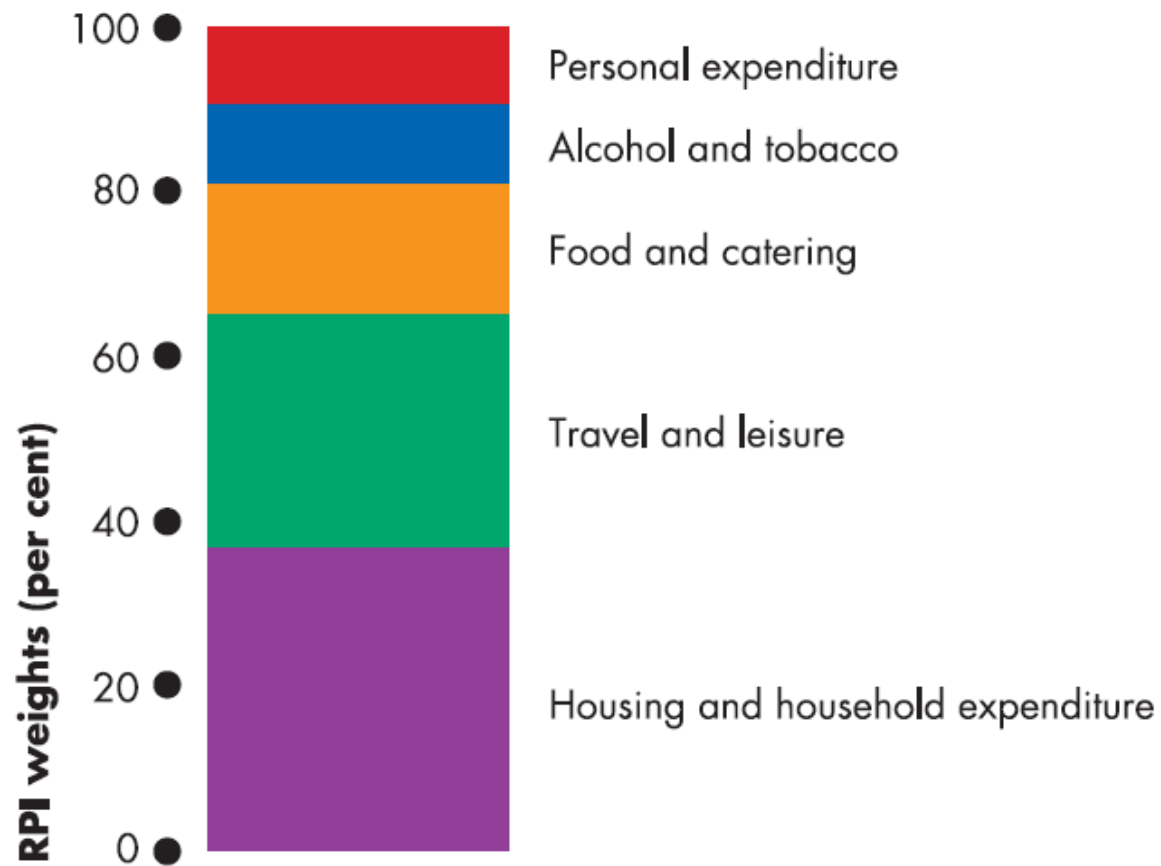
The for the average of the 36 months from January 1982 through December 1984 in the US, the CPI equals 100

In June 2010, the CPI was 218, meaning the average of the prices paid by urban consumers for a fixed market basket of consumer goods and services was 118 percent higher

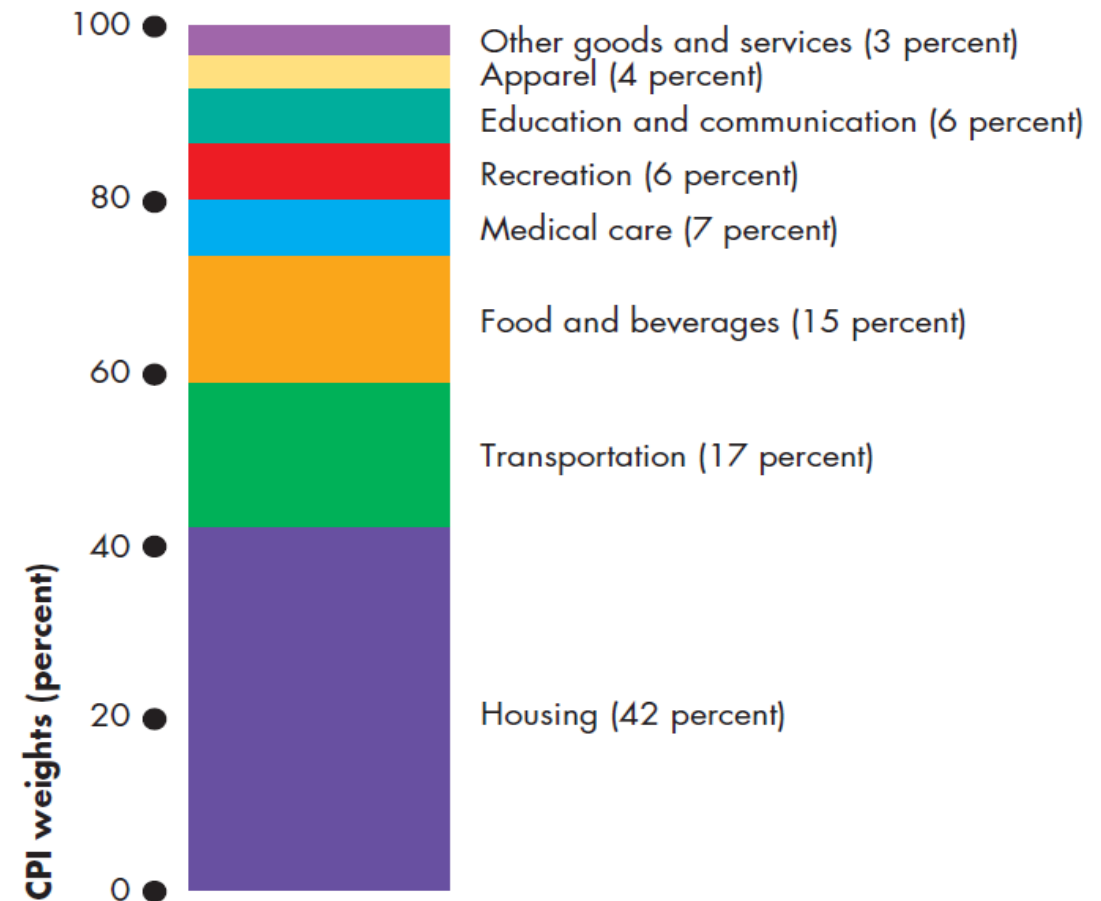
Construction and Calculation of CPI/RPI

- ***Selecting the appropriate basket:*** The first stage is to construct an appropriate basket of goods and services, weighted by their relative importance
- ***Conducting monthly price surveys:*** Monthly surveys are conducted to record the prices of the exact goods and services from the “basket”
- ***Calculating the CPI/RPI:*** After the surveys are conducted the following steps are done to calculate the CPI/RPI.
 - Cost of the basket at base-period prices
 - Cost of the CPI basket at current-period prices
 - Calculate the index for the base period and the current period

Construction and Calculation of CPI/RPI



(a) The RPI basket



(b) The CPI basket

Construction and Calculation of CPI/RPI

- Formula for calculating CPI:

$$CPI = \frac{\text{Cost of CPI basket at current prices}}{\text{Cost of CPI basket at base period prices}} \times 100$$

- Formula for calculating RPI:

$$RPI = \frac{\text{Cost of basket at current prices}}{\text{Cost of basket at base period prices}} \times 100$$

Example

- Calculate the CPI for the 2010, 2011 and 2012 from the following information. Assume 2010 to be the base-period.

Item	Quantity	Price in 2010 (US\$)	Price in 2011 (US\$)	Price in 2012 (US\$)
Oranges	10	1.00	2.00	1.50
Haircuts	5	8.00	10.00	9.00

Measuring Inflation

- Primary purpose of calculating CPI is to measure the changes in cost of living and the value of money.
- The inflation rate measures this changes in the cost of living and the value of money
- The inflation rate is calculated by measuring the annual percentage change in the CPI

$$\text{Inflation rate} = \frac{\text{CPI this year} - \text{CPI last year}}{\text{CPI last year}} \times 100$$

Example

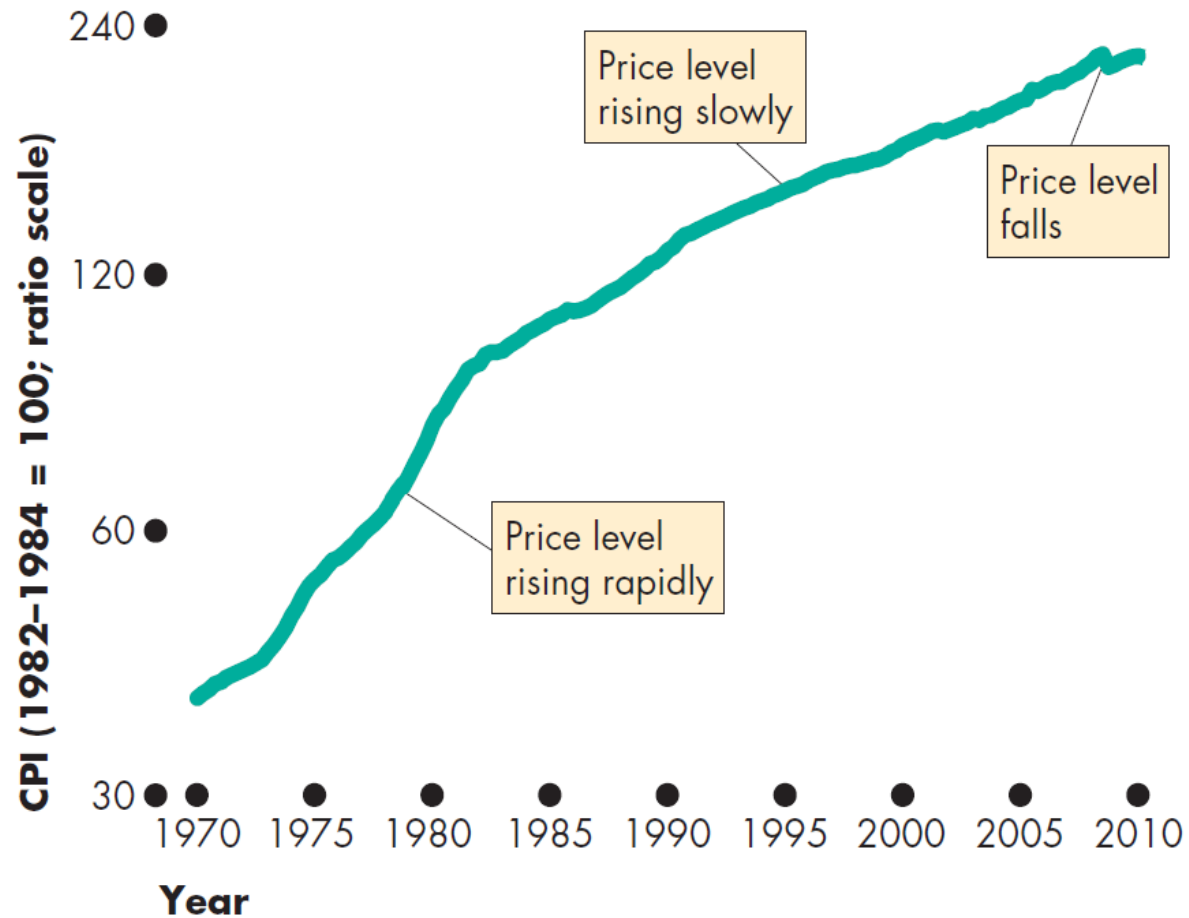
- The BLS reported the following CPI data. Calculate the inflation rate for the years ended in June 2024 and June 2025.

Year	CPI
June 2023	201.9
June 2024	207.2
June 2025	217.4

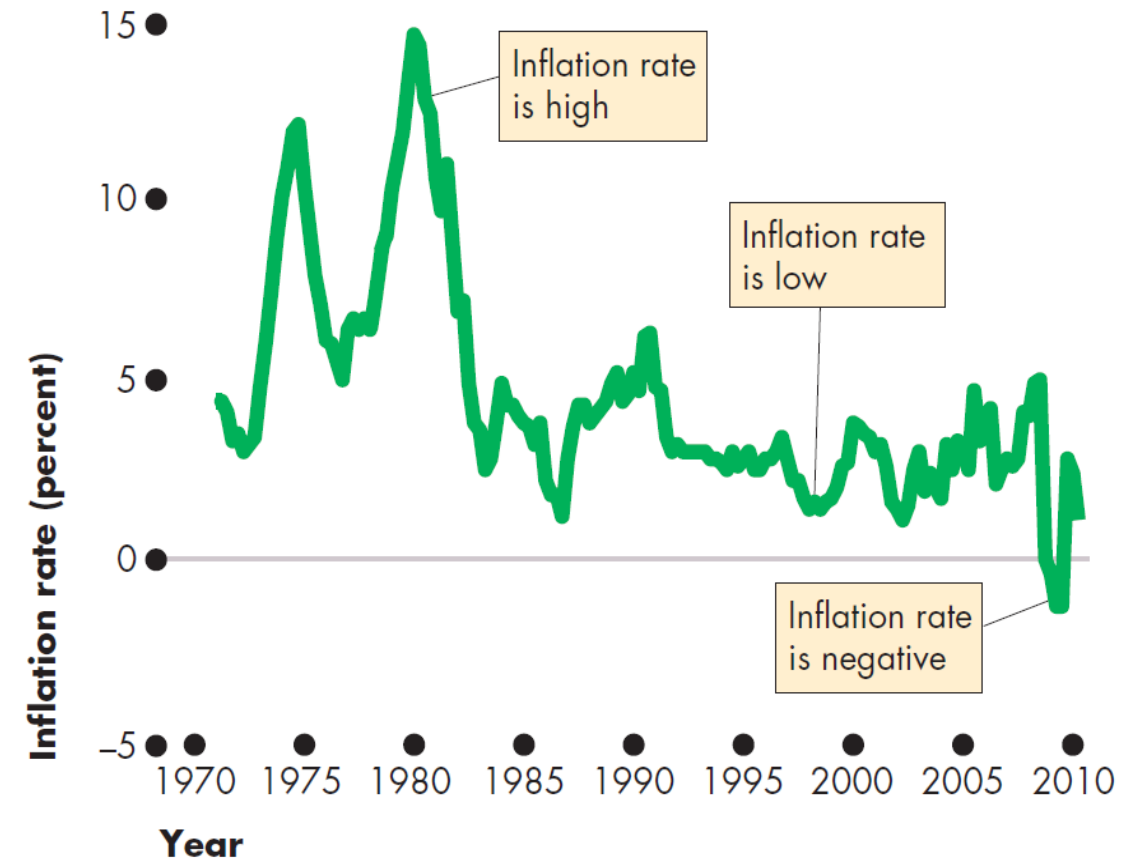
High Price Level & High Inflation

- When the price level ***rises rapidly***, the inflation rate is ***high***
- When the price level ***rises slowly***, the inflation rate is ***low***
- When the price level ***falls***, the inflation rate is ***negative*** (deflation)

High Price Level & High Inflation



(a) CPI



(b) Inflation rate

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Biasness in CPI

- ***New Goods Bias:*** New goods puts and upward bias into the CPI and its inflation rate
- ***Quality Change Bias:*** CPI considers the price rise of product due to improved quality as inflation and overstates it
- ***Commodity Substitution Bias:*** CPI fails to take the substitution of a product into account
- ***Outlet Substitution Bias:*** The CPI surveys do not monitor outlet substitutions, the substitution from convenience stores to discount stores when faced with higher prices

Question

IF INFLATION MEANS PRICES ARE RISING, WHY DO PEOPLE STILL SPEND MORE DURING INFLATIONARY PERIODS INSTEAD OF CUTTING BACK?